

STATEMENT OF WORK

FOR

**B1005 & B1007 CE Compound Gas Valve
Replacement**

606 S. Aspen St.
Buckley SFB, Colorado 80011

COA: 1 April 2026

1.0 Scope of Work

1.1 Primary Objective

The contractor shall provide fully qualified technicians and all specialized tools to replace six (6) gas control valves suspected of damage from a faulty gas regulator.

1.2 Gas valve install, System Inspections and Re-commissioning

The contractor shall thoroughly review all equipment's manufacturer Installation and Operation manuals in their entirety prior to beginning any work. After reviewing all manuals, the contractor will inspect and carefully install all gas valves according to manufacturer's specifications. Extra precautions will be taken, and the following will be avoided to prevent potential damage to new gas valves or systems.

- Cross threading any connections or fittings.
- Using excessive pipe sealant that can restrict gas flow.
- Applying excessive force directly to the new gas valve assembly or body while installing.

Following valve replacement, the contractor shall:

- Conduct comprehensive leak checks on all affected equipment.
- Perform a full safety inspection of all gas-fired equipment, including internal components such as burners, gas valves, heat exchangers, and all electrical/mechanical parts.
- Perform multiple leak checks on each piece of equipment, both while the units are static and again while under full operational load.
- Re-commission all boilers and water heaters after a full operational test to certify their continued safe operation.

1.3 Deficiency Reporting

Upon completion of the re-commissioning, the contractor will deliver a deficiency report detailing any compromised components or units requiring further attention. If additional work or a full system replacement is recommended, the contractor must provide a separate, detailed report with clear justification for the recommendation.

1.4 Contractor-Furnished Items

The contractor is responsible for providing all necessary materials and equipment, including but not limited to:

- Gas control valves, assemblies, fittings, gaskets, and valve replacement kits.
- All specialized tools required to re-commission units, perform gas pressure tests, conduct combustion analysis, and complete operational tests of all gas-fired equipment.

1.5 Governing Standards

All work shall be performed in strict accordance with the manufacturer's specifications and the latest editions of the following codes and standards:

- UFC 3-410-01: Heating, Ventilating, and Air Conditioning Systems
- UFC 3-420-01: Plumbing Systems
- ASME Boiler and Pressure Vessel Codes
- All applicable state and local laws/codes

UFC criteria can be accessed via the Whole Building Design Guide at <https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc>.

2.0 Project Logistics & Scheduling

- **Period of Performance:** 60 days after Receipt of Award
 - **Building Downtime:** No specific building-wide downtime is required. Individual units can be isolated for work as needed, ensuring continuous operations.
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3.0 Technical & Compliance Requirements

3.1 Definition of Qualified Labor

The contractor must ensure all technicians are certified and licensed to service and repair commercial boilers in accordance with the standards listed in Section 1.5. Furthermore, all technicians must possess specialized certification and experience with the following specific equipment models:

- **Rinnai Tankless Water Heaters:** Model # REU-KBD3237FFUDC-US • **Lochinvar Knight Boilers:** Model # KBN601 • **Lochinvar Knight Boilers:** Model # KBN286

3.2 System Pressure Testing Protocol

The contractor shall verify the maximum and minimum inlet gas pressures for each unit via its data plate before commencing work. Inlet pressure tests must be performed before and after new valve installation. **Under no circumstances shall any valve be exposed to pressures exceeding the manufacturer's maximum inlet pressure specification.** After installation, the contractor will verify the working inlet gas pressure with all systems running to ensure valves are functioning within manufacturer specifications.

3.3 Reporting & Deliverables

Upon project completion, the contractor shall submit a final report containing the following detailed information:

- Pre- and post-installation gas pressure readings.
- Results of all leak checks (for both static and operational states).
- A summary of all operational and safety checks performed.
- A complete combustion analysis report.
- A log of all replaced parts.
- A detailed summary of any unit deficiencies discovered during re-commissioning.

3.4 Return of Materials

The contractor shall tag all removed parts with the corresponding unit they were removed from and return them directly to the 460 CES project manager.

Attachments

- Attachment 1: Rinnai data plate photo (2)
- Attachment 2: Lochinvar data plate photo (KBN601) (3)
- Attachment 3: Lochinvar data plate photo (KBN286) (1)
- Attachment 4: Lochinvar KBN601 O & I (preview)
- Attachment 5: Lochinvar KBN286 O & I (preview)

ATTACHMENT 1

Direct-Vent Automatic Instantaneous Water Heater

Chauffe-eau instantané automatique à ventilation directe

Type of Gas / (Type de gaz) : **Natural Gas (Gaz naturel)**

Input Rating / (Débit calorifique) : **Maximum : 199,000 BTU/H**
Minimum : 15,200 BTU/H

Maximum Inlet Gas Pressure / Maximum à l'arrivée d'alimentation : **10.5" W.C. / (2.61 kPa)**

Minimum Inlet Gas Pressure / Minimum à l'arrivée d'alimentation : **4.0" W.C. / (1.00 kPa)**

Minimum Inlet Gas press for adjustments / Minimum (pour les réglages) : **7.0" W.C. / (1.74 kPa)**

Manifold Pressure – High Fire / Pression à la rampe (plein feu) : **3.0" W.C. / (0.75 kPa)**

Manifold Pressure – Low Fire / Pression à la rampe (petit feu) : **0.42" W.C. / (0.10 kPa)**

Recovery Rating / Taux de récupération (GPH) : **234**

Maximum Water Pressure / Pression d'eau maximum : **150 psi**

NOx Emissions : **< 14 ng/J**

(Pursuant to South Coast Air Quality Management District Rule 1146.2)
 (suivant la réglementation de district en côte Sud sur la gestion de la qualité de l'air)

Minimum clearances from combustibles / Dégagements minimaux à assurer entre les parois de l'appareil et les constructions combustibles :

Back/Arrière : 0	Front/Avant : 6" (15.2cm)	Top/Dessus : 6" (15.2cm)
Sides/Côtés : 2" (5.1cm)	Ground/Sol : 12" (30.5cm)	Vent/Ventilation : 0

For Closet Installations, 6" from front / Pour installation dans un endroit fermé : 6" de l'avant

Minimum clearances from non-combustibles / Dégagements minimaux à assurer entre les parois de l'appareil et les constructions incombustibles :

Back/Arrière : 0	Front/Avant : 6" (15.2cm)	Top/Dessus : 2" (5.1cm)
Sides/Côtés : 1/2" (1.3cm)	Ground/Sol : 12" (30.5cm)	Vent/Ventilation : 0

Rinnai

CORPORATION
Nagoya, Japan

FOR INDOOR INSTALLATION ONLY / POUR INSTALLATION EN INTÉRIEUR SEULEMENT

120 VOLT 60 HZ less than 10 AMPS / 120 V 60 Hz, moins de 10 A

For field-wired connections use No. 14 AWG wires rated for at least 140°F (60°C) /
 Pour les connexions sur site utilisez des fils de calibre 14 AWG supportant au moins 140°F (60°C).

Minimum Wall Thickness / Épaisseur minimale de mur : 4 Inches (10.2 cm)
 Maximum Wall Thickness / Épaisseur maximale de mur : 20 Inches (51 cm)

ANSI Z21. 10. 3-2014
/ CSA 4. 3-2014

Model #: C199i
(REU-KBD3237FFUDC-US)

SERIAL NUMBER
GF. BA-032726

MADE IN JAPAN
US Pat. D533, 928 S

This equipment complies with the 2001 requirements of ASHRAE Standard 90. 1.

Suitable for water (potable) heating and space heating / Pour chauffage de l'eau (potable) et des locaux

WARNING!!

DO NOT USE TOXIC CHEMICALS, SUCH AS ARE USED FOR BOILER TREATMENT IN POTABLE WATER HEATING SYSTEMS USED FOR SPACE HEAT.

AVERTISSEMENT!!

N'UTILISEZ PAS DE PRODUITS CHIMIQUES, COMME CEUX SERVANT POUR LE TRAITEMENT DE CHAUDIÈRES, DANS LES SYSTÈMES DE CHAUFFAGE D'EAU POTABLE UTILISÉS AUSSI EN CHAUFFAGE CENTRAL.

23-2872

ATTACHMENT 2



[illegible]



7 Gas connections *(continued)*

⚠ WARNING When re-tightening the set screw, be sure to tighten securely to prevent gas leaks.

Do not check for gas leaks with an open flame -- use the bubble test. Failure to use the bubble test or check for gas leaks can cause severe personal injury, death, or substantial property damage.

14. Turn on the gas supply at the manual gas valve.
15. Turn the power switch to the "ON" position.
16. Adjust the temperature set point on the control panel of the SMART SYSTEM control module to the desired water temperature so the appliance will call for heat.
17. Check burner performance by cycling the system while you observe burner response. The burner should ignite promptly. Flame pattern should be stable. Turn system off and allow burner to cool, then cycle burner again to ensure proper ignition and flame characteristics.

Figure 7-5 Inlet Gas Supply Check - Model 400

LOOSEN THE SET SCREW ONE (1) FULL TURN AND PLACE THE MANOMETER TUBING OVER THE PRESSURE TAP.

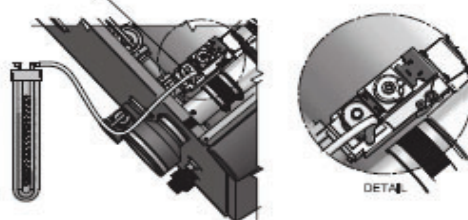


Figure 7-6 Inlet Gas Supply Check - Model 501

LOOSEN THE SET SCREW ONE (1) FULL TURN AND PLACE THE MANOMETER TUBING OVER THE PRESSURE TAP.

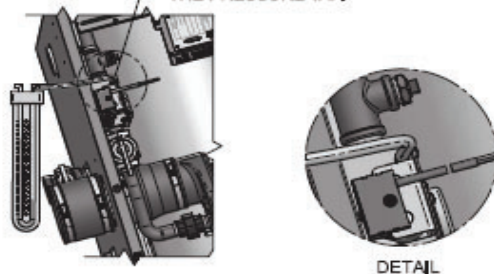
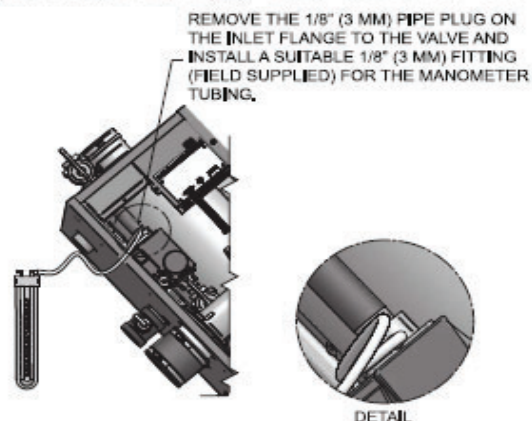


Figure 7-7 Inlet Gas Supply Check - Models 601 - 801



Gas pressure

The gas pressure must remain between 4 inches w.c. (.99 kPa) minimum and 14 inches w.c. (3.5 kPa) maximum for Natural gas and between 8 inches w.c. (1.9 kPa) minimum and 14 inches w.c. (3.2 kPa) maximum for LP gas during standby (static) mode and while in operating (dynamic) mode. If an in-line regulator is used, it must be a minimum of 10 feet (3 m) from the Knight XL boiler. It is very important that the gas line is properly purged by the gas supplier or utility company. Failure to properly purge the lines or improper line sizing, will result in ignition failure.

The problem is especially noticeable in NEW LP installations and also in empty tank situations. This can also occur when a utility company shuts off service to an area to provide maintenance to their lines.

Gas valve replacement

The gas valve **MUST NOT** be replaced with a conventional gas valve under any circumstances. As an additional safety feature, this gas valve has a flanged connection to the venturi and blower.

⚠ WARNING Failure to follow all precautions could result in fire, explosion, or death!

⚠ WARNING DO NOT adjust gas valve outlet pressure. Attempting to alter the gas valve outlet pressure could result in damage to the valve, causing potential severe personal injury, death, or substantial property damage.



7 Gas connections *(continued)*

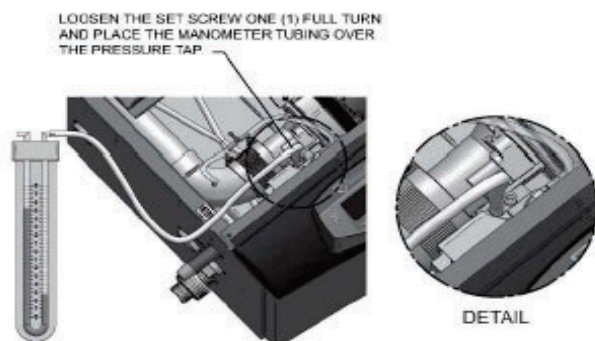
⚠ WARNING

When re-tightening the set screw, be sure to tighten securely to prevent gas leaks.

Do not check for gas leaks with an open flame -- use the bubble test. Failure to use the bubble test or check for gas leaks can cause severe personal injury, death, or substantial property damage.

14. Turn on the gas supply at the manual gas valve.
15. Turn the power switch to the "ON" position.
16. Adjust the temperature set point on the control panel of the SMART SYSTEM control module to the desired water temperature so the appliance will call for heat.
17. Check burner performance by cycling the system while you observe burner response. The burner should ignite promptly. Flame pattern should be stable. Turn system off and allow burner to cool, then cycle burner again to ensure proper ignition and flame characteristics.

Figure 7-4 Inlet Gas Supply Check



Gas Pressure

The gas pressure must remain between 4 inches w.c. (natural), 8 inches w.c. (LP) minimum and 14 inches w.c. (natural and LP) maximum during stand-by (static) mode and while in operating (dynamic) mode. If an in-line regulator is used, it must be a minimum of 10 feet from the Knight boiler. It is very important that the gas line is properly purged by the gas supplier or utility company. Failure to properly purge the lines or improper line sizing, will result in ignition failure.

The problem is especially noticeable in NEW LP installations and also in empty tank situations. This can also occur when a utility company shuts off service to an area to provide maintenance to their lines.

Gas valve replacement

The gas valve **MUST NOT** be replaced with a conventional gas valve under any circumstances. As an additional safety feature, this gas valve has a flanged connection to the venturi and blower.

⚠ WARNING

Failure to follow all precautions could result in fire, explosion, or death!

⚠ WARNING

DO NOT adjust gas valve outlet pressure. The gas valve is factory-set for the correct outlet pressure. This setting is suitable for natural gas and propane, requiring no field adjustment. Attempting to alter the gas valve outlet pressure could result in damage to the valve, causing potential severe personal injury, death, or substantial property damage.